

# Mathematics Extension 2

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**General  
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

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**Total marks:  
100****Section I – 10 marks** (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

**Section II – 90 marks** (pages 7–18)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

## Section I

10 marks

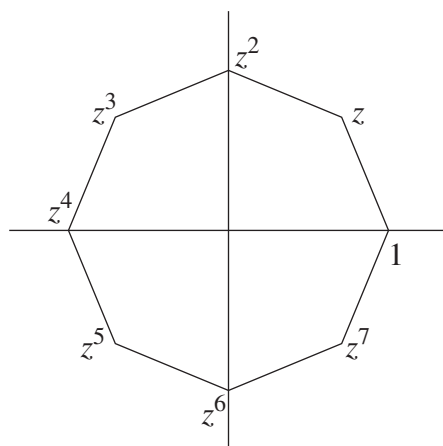
Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

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- 1 The complex number  $z$  is chosen so that  $1, z, \dots, z^7$  form the vertices of the regular polygon shown.



Which polynomial equation has all of these complex numbers as roots?

- A.  $x^7 - 1 = 0$
  - B.  $x^7 + 1 = 0$
  - C.  $x^8 - 1 = 0$
  - D.  $x^8 + 1 = 0$
- 2 Suppose  $\ell$  is a line and  $S$  is a point NOT on  $\ell$ .

The point  $P$  moves so that the distance from  $P$  to  $S$  is half the perpendicular distance from  $P$  to  $\ell$ .

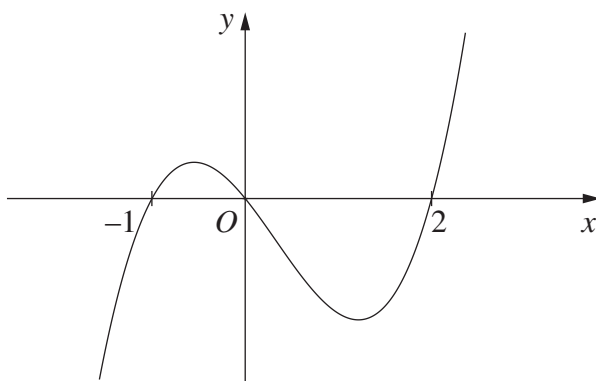
Which conic best describes the locus of  $P$ ?

- A. A circle
- B. An ellipse
- C. A parabola
- D. A hyperbola

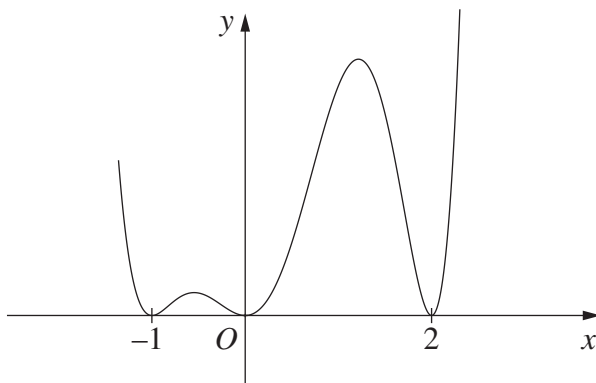
3 Which complex number lies in the region  $2 < |z - 1| < 3$ ?

- A.  $1 + \sqrt{3}i$
- B.  $1 + 3i$
- C.  $2 + i$
- D.  $3 - i$

4 The graph of the function  $y = f(x)$  is shown.



A second graph is obtained from the function  $y = f(x)$ .



Which equation best represents the second graph?

- A.  $y = \sqrt{f(x)}$
- B.  $y^2 = f(x)$
- C.  $y = [f(x)]^2$
- D.  $y = f(x^2)$

- 5 The polynomial  $p(x) = x^3 - 2x + 2$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

What is the value of  $\alpha^3 + \beta^3 + \gamma^3$ ?

- A.  $-10$
- B.  $-6$
- C.  $-2$
- D.  $0$

- 6 It is given that  $z = 2 + i$  is a root of  $z^3 + az^2 - 7z + 15 = 0$ , where  $a$  is a real number.

What is the value of  $a$ ?

- A.  $-1$
- B.  $1$
- C.  $7$
- D.  $-7$

- 7 It is given that  $f(x)$  is a non-zero even function and  $g(x)$  is a non-zero odd function.

Which expression is equal to  $\int_{-a}^a f(x) + g(x) dx$ ?

- A.  $2 \int_0^a f(x) dx$
- B.  $2 \int_0^a g(x) dx$
- C.  $\int_{-a}^a g(x) dx$
- D.  $2 \int_0^a f(x) + g(x) dx$

- 8 Suppose that  $f(x)$  is a non-zero odd function.

Which of the functions below is also odd?

- A.  $f(x^2)\cos x$
- B.  $f(f(x))$
- C.  $f(x^3)\sin x$
- D.  $f(x^2) - f(x)$

- 9 A particle is travelling on the circle with equation  $x^2 + y^2 = 16$ .

It is given that  $\frac{dx}{dt} = y$ .

Which statement about the motion of the particle is true?

- A.  $\frac{dy}{dt} = x$  and the particle travels clockwise
- B.  $\frac{dy}{dt} = x$  and the particle travels anticlockwise
- C.  $\frac{dy}{dt} = -x$  and the particle travels clockwise
- D.  $\frac{dy}{dt} = -x$  and the particle travels anticlockwise

- 10 Suppose  $f(x)$  is a differentiable function such that

$$\frac{f(a)+f(b)}{2} \geq f\left(\frac{a+b}{2}\right), \text{ for all } a \text{ and } b.$$

Which statement is always true?

A.  $\int_0^1 f(x) dx \geq \frac{f(0)+f(1)}{2}$

B.  $\int_0^1 f(x) dx \leq \frac{f(0)+f(1)}{2}$

C.  $f'\left(\frac{1}{2}\right) \geq 0$

D.  $f'\left(\frac{1}{2}\right) \leq 0$

## Section II

**90 marks**

**Attempt Questions 11–16**

**Allow about 2 hours and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

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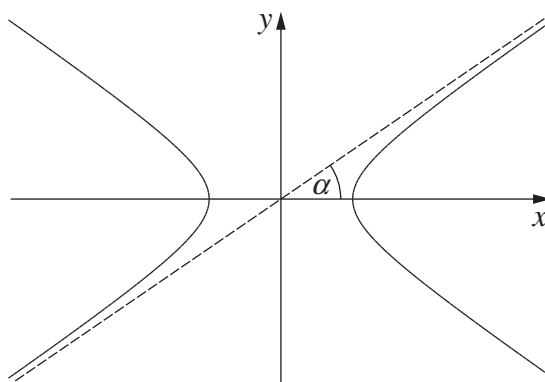
**Question 11** (15 marks) Use a SEPARATE writing booklet.

(a) Let  $z = 1 - \sqrt{3}i$  and  $w = 1 + i$ .

(i) Find the exact value of the argument of  $z$ . **1**

(ii) Find the exact value of the argument of  $\frac{z}{w}$ . **2**

(b) An asymptote to the hyperbola  $\frac{x^2}{12} - \frac{y^2}{4} = 1$  makes an angle  $\alpha$  with the positive  $x$ -axis, as shown. **2**



Find the value of  $\alpha$ .

(c) Sketch the region in the Argand diagram where **2**

$$-\frac{\pi}{4} \leq \arg z \leq 0 \quad \text{and} \quad |z - 1 + i| \leq 1.$$

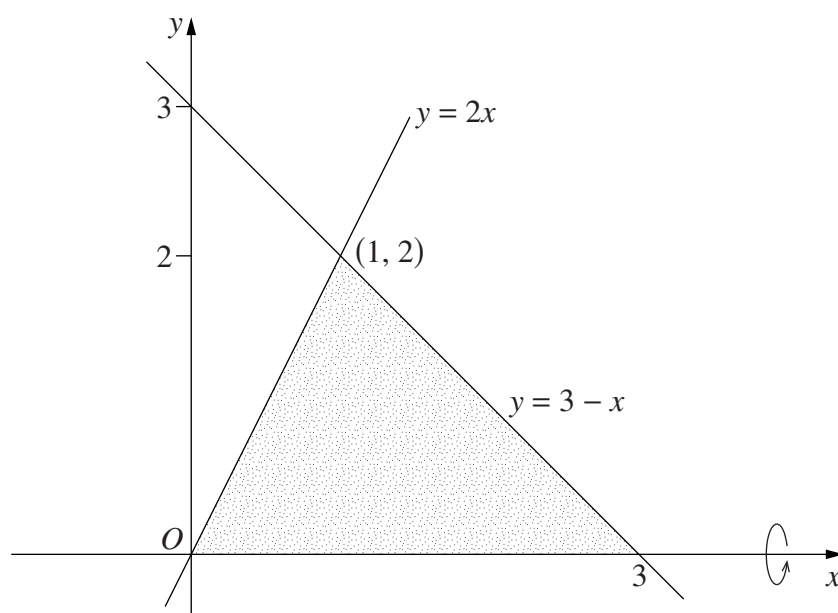
**Question 11 continues on page 8**

Question 11 (continued)

- (d) Using the substitution  $t = \tan \frac{\theta}{2}$ , or otherwise, evaluate **3**

$$\int_0^{\frac{2\pi}{3}} \frac{1}{1 + \cos \theta} d\theta.$$

- (e) The region bounded by the lines  $y = 3 - x$ ,  $y = 2x$  and the  $x$ -axis is rotated about the  $x$ -axis. **2**



Use the method of cylindrical shells to find an integral whose value is the volume of the solid of revolution formed. Do NOT evaluate the integral.

- (f) Using the substitution  $x = \sin^2 \theta$ , or otherwise, evaluate  $\int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} dx$ . **3**

**End of Question 11**

**Question 12** (15 marks) Use a SEPARATE writing booklet.

(a) Consider the function  $f(x) = \frac{e^x - 1}{e^x + 1}$ .

(i) Show that  $f(x)$  is increasing for all  $x$ . **1**

(ii) Show that  $f(x)$  is an odd function. **1**

(iii) Describe the behaviour of  $f(x)$  for large positive values of  $x$ . **1**

(iv) Hence sketch the graph of  $f(x) = \frac{e^x - 1}{e^x + 1}$ . **1**

(v) Hence, or otherwise, sketch the graph of  $y = \frac{1}{f(x)}$ . **1**

(b) Solve the quadratic equation  $z^2 + (2 + 3i)z + (1 + 3i) = 0$ , giving your answers in the form  $a + bi$ , where  $a$  and  $b$  are real numbers. **3**

(c) Find  $\int x \tan^{-1} x \, dx$ . **3**

(d) Let  $P(x)$  be a polynomial.

(i) Given that  $(x - \alpha)^2$  is a factor of  $P(x)$ , show that **2**

$$P(\alpha) = P'(\alpha) = 0.$$

(ii) Given that the polynomial  $P(x) = x^4 - 3x^3 + x^2 + 4$  has a factor  $(x - \alpha)^2$ , find the value of  $\alpha$ . **2**

**Question 13** (15 marks) Use a SEPARATE writing booklet.

(a) Show that  $\frac{r+s}{2} \geq \sqrt{rs}$  for  $r \geq 0$  and  $s \geq 0$ . **1**

(b) Let  $a, b$  and  $c$  be real numbers. Suppose that  $P(x) = x^4 + ax^3 + bx^2 + cx + 1$  has roots  $\alpha, \frac{1}{\alpha}, \beta, \frac{1}{\beta}$ , where  $\alpha > 0$  and  $\beta > 0$ .

(i) Prove that  $a = c$ . **2**

(ii) Using the inequality in part (a), show that  $b \geq 6$ . **2**

(c) A particle is projected upwards from ground level with initial velocity  $\frac{1}{2}\sqrt{\frac{g}{k}} \text{ m s}^{-1}$ , where  $g$  is the acceleration due to gravity and  $k$  is a positive constant. The particle moves through the air with speed  $v \text{ m s}^{-1}$  and experiences a resistive force. **4**

The acceleration of the particle is given by  $\ddot{x} = -g - kv^2 \text{ m s}^{-2}$ . Do NOT prove this.

The particle reaches a maximum height,  $H$ , before returning to the ground.

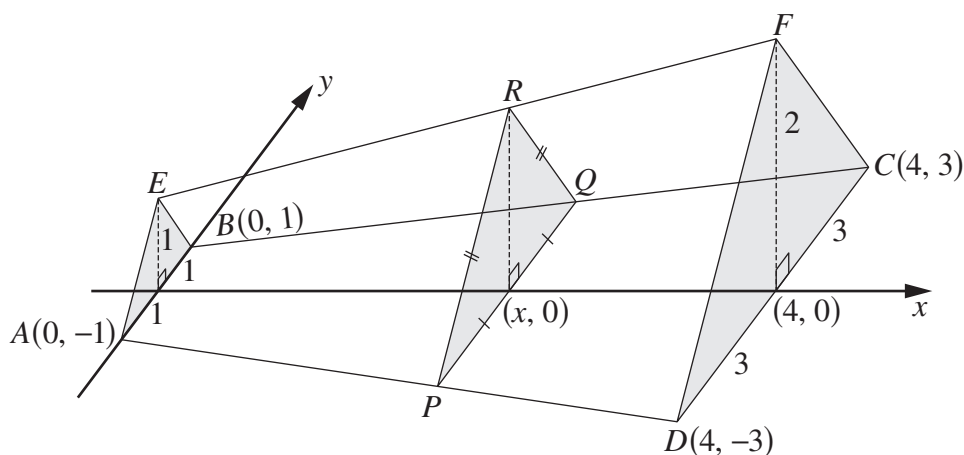
Using  $\ddot{x} = v \frac{dv}{dx}$ , or otherwise, show that  $H = \frac{1}{2k} \log_e \left( \frac{5}{4} \right)$  metres.

**Question 13 continues on page 11**

Question 13 (continued)

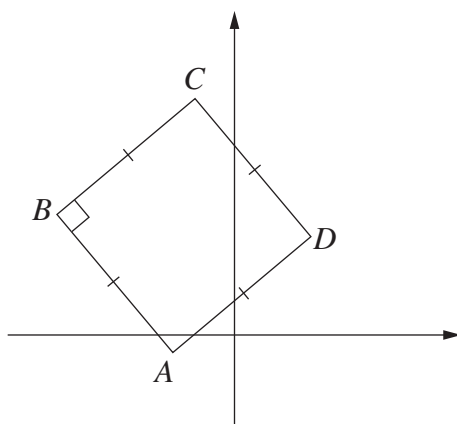
- (d) The trapezium whose vertices are  $A(0, -1)$ ,  $B(0, 1)$ ,  $C(4, 3)$  and  $D(4, -3)$  forms the base of a solid. 4

Each cross-section perpendicular to the  $x$ -axis is an isosceles triangle. The height of the isosceles triangle  $ABE$  with base  $AB$ , is 1. The height of the isosceles triangle  $DCF$  with base  $DC$ , is 2. The cross-section through the point  $(x, 0)$  is the isosceles triangle  $PQR$ , where  $R$  lies on the line  $EF$ , as shown in the diagram.



Find the volume of the solid.

- (e) The points  $A$ ,  $B$ ,  $C$  and  $D$  on the Argand diagram represent the complex numbers  $a$ ,  $b$ ,  $c$  and  $d$  respectively. The points form a square as shown on the diagram. 2



By using vectors, or otherwise, show that  $c = (1 + i)d - ia$ .

**End of Question 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet.

(a) It is given that

$$x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2).$$

(i) Find  $A$  and  $B$  so that

**1**

$$\frac{16}{x^4 + 4} = \frac{A + 2x}{x^2 + 2x + 2} + \frac{B - 2x}{x^2 - 2x + 2}.$$

(ii) Hence, or otherwise, show that for any real number  $m$ ,

**2**

$$\int_0^m \frac{16}{x^4 + 4} dx = \ln\left(\frac{m^2 + 2m + 2}{m^2 - 2m + 2}\right) + 2 \tan^{-1}(m + 1) + 2 \tan^{-1}(m - 1).$$

(iii) Find the limiting value as  $m \rightarrow \infty$  of

**1**

$$\int_0^m \frac{16}{x^4 + 4} dx.$$

**Question 14 continues on page 13**

Question 14 (continued)

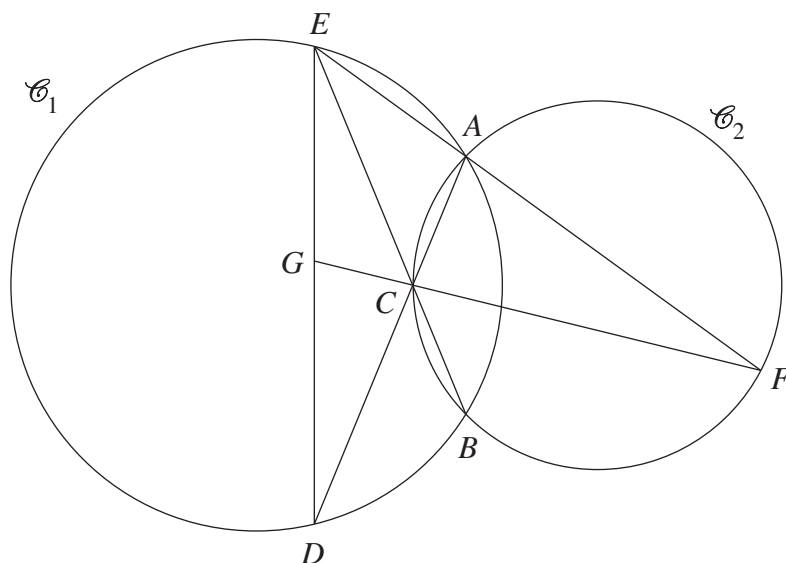
- (b) Two circles,  $\mathcal{C}_1$  and  $\mathcal{C}_2$ , intersect at the points  $A$  and  $B$ . Point  $C$  is chosen on the arc  $AB$  of  $\mathcal{C}_2$  as shown in the diagram.

The line segment  $AC$  produced meets  $\mathcal{C}_1$  at  $D$ .

The line segment  $BC$  produced meets  $\mathcal{C}_1$  at  $E$ .

The line segment  $EA$  produced meets  $\mathcal{C}_2$  at  $F$ .

The line segment  $FC$  produced meets the line segment  $ED$  at  $G$ .



Copy or trace the diagram into your writing booklet.

- |                                                                              |          |
|------------------------------------------------------------------------------|----------|
| (i) State why $\angle EAD = \angle EBD$ .                                    | <b>1</b> |
| (ii) Show that $\angle EDA = \angle AFC$ .                                   | <b>1</b> |
| (iii) Hence, or otherwise, show that $B, C, G$ and $D$ are concyclic points. | <b>3</b> |

**Question 14 continues on page 14**

Question 14 (continued)

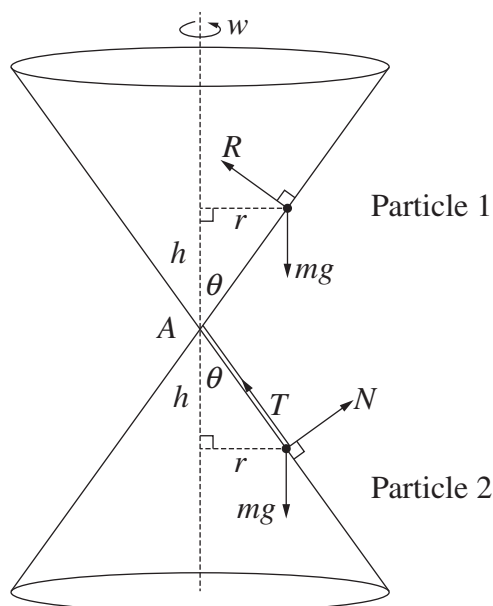
- (c) A smooth double cone with semi-vertical angle  $\theta < \frac{\pi}{2}$  is rotating about its axis with constant angular velocity  $w$ .

Two particles, each of mass  $m$ , are sitting on the cone as it rotates, as shown in the diagram.

Particle 1 is inside the cone at vertical distance  $h$  above the apex,  $A$ , and moves in a horizontal circle of radius  $r$ .

Particle 2 is attached to the apex  $A$  by a light inextensible string so that it sits on the cone at vertical distance  $h$  below the apex. Particle 2 also moves in a horizontal circle of radius  $r$ .

The acceleration due to gravity is  $g$ .



- (i) The normal reaction force on Particle 1 is  $R$ . 2

By resolving  $R$  into vertical and horizontal components, or otherwise, show that  $w^2 = \frac{gh}{r^2}$ .

- (ii) The normal reaction force on Particle 2 is  $N$  and the tension in the string is  $T$ . 2

By considering horizontal and vertical forces, or otherwise, show that

$$N = mg \left( \sin \theta - \frac{h}{r} \cos \theta \right).$$

- (iii) Show that  $\theta \geq \frac{\pi}{4}$ . 2

**End of Question 14**

**Question 15** (15 marks) Use a SEPARATE writing booklet.

(a) Let  $I_n = \int_0^1 x^n \sqrt{1-x^2} dx$ , for  $n = 0, 1, 2, \dots$ .

(i) Find the value of  $I_1$ . **1**

(ii) Using integration by parts, or otherwise, show that for  $n \geq 2$  **3**

$$I_n = \left( \frac{n-1}{n+2} \right) I_{n-2}.$$

(iii) Find the value of  $I_5$ . **1**

(b) Consider the curve  $\sqrt{x} + \sqrt{y} = \sqrt{a}$ , for  $x \geq 0$  and  $y \geq 0$ , where  $a$  is a positive constant.

(i) Show that the equation of the tangent to the curve at the point  $P(c, d)$  is given by  $y\sqrt{c} + x\sqrt{d} = d\sqrt{c} + c\sqrt{d}$ . **2**

(ii) The tangent to the curve at the point  $P$  meets the  $x$  and  $y$  axes at  $A$  and  $B$  respectively. Show that  $OA + OB = a$ , where  $O$  is the origin. **3**

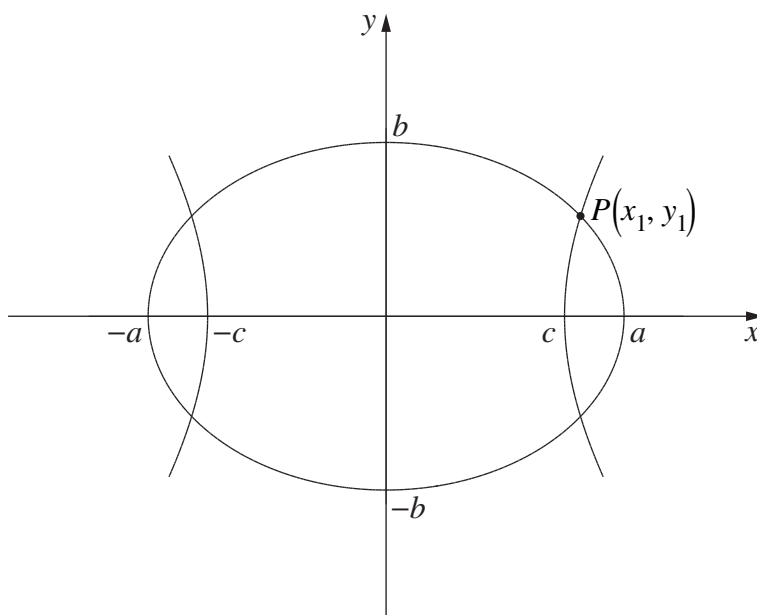
**Question 15 continues on page 16**

Question 15 (continued)

- (c) The ellipse with equation  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ , where  $a > b$ , has eccentricity  $e$ .

The hyperbola with equation  $\frac{x^2}{c^2} - \frac{y^2}{d^2} = 1$ , has eccentricity  $E$ .

The value of  $c$  is chosen so that the hyperbola and the ellipse meet at  $P(x_1, y_1)$ , as shown in the diagram.



- (i) Show that  $\frac{x_1^2}{y_1^2} = \frac{a^2 c^2}{(a^2 - c^2)} \times \frac{(b^2 + d^2)}{b^2 d^2}$ . 2
- (ii) If the two conics have the same foci, show that their tangents at  $P$  are perpendicular. 3

**End of Question 15**

**Question 16** (15 marks) Use a SEPARATE writing booklet.

(a) Let  $\alpha = \cos \theta + i \sin \theta$ , where  $0 < \theta < 2\pi$ .

(i) Show that  $\alpha^k + \alpha^{-k} = 2 \cos k\theta$ , for any integer  $k$ . 1

Let  $C = \alpha^{-n} + \cdots + \alpha^{-1} + 1 + \alpha + \cdots + \alpha^n$ , where  $n$  is a positive integer.

(ii) By summing the series, prove that  $C = \frac{\alpha^n + \alpha^{-n} - (\alpha^{n+1} + \alpha^{-(n+1)})}{(1 - \alpha)(1 - \bar{\alpha})}$ . 3

(iii) Deduce, from parts (i) and (ii), that 2

$$1 + 2(\cos \theta + \cos 2\theta + \cdots + \cos n\theta) = \frac{\cos n\theta - \cos(n+1)\theta}{1 - \cos \theta}.$$

(iv) Show that  $\cos \frac{\pi}{n} + \cos \frac{2\pi}{n} + \cdots + \cos \frac{n\pi}{n}$  is independent of  $n$ . 1

(b) The hyperbola with equation 2

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

has eccentricity 2.

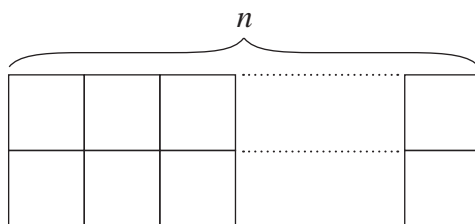
The distance from one of the foci to one of the vertices is 1.

What are the possible values of  $a$ ?

**Question 16 continues on page 18**

Question 16 (continued)

- (c) A 2 by  $n$  grid is made up of two rows of  $n$  square tiles, as shown.



The tiles of the 2 by  $n$  grid are to be painted so that tiles sharing an edge are painted using different colours. There are  $x$  different colours available, where  $x \geq 2$ .

It is NOT necessary to use all the colours.

Consider the case of the 2 by 2 grid with tiles labelled  $A$ ,  $B$ ,  $C$  and  $D$ , as shown.

|     |     |
|-----|-----|
| $A$ | $C$ |
| $B$ | $D$ |

There are  $x(x - 1)$  ways to choose colours for the first column containing tiles  $A$  and  $B$ . Do NOT prove this.

- (i) Assume the colours for tiles  $A$  and  $B$  have been chosen. There are two cases to consider when choosing colours for the second column. Either tile  $C$  is the same colour as tile  $B$ , or tile  $C$  is a different colour from tile  $B$ . 2

By considering these two cases, show that the number of ways of choosing colours for the second column is

$$x^2 - 3x + 3.$$

- (ii) Prove by mathematical induction that the number of ways in which the 2 by  $n$  grid can be painted is  $x(x - 1)(x^2 - 3x + 3)^{n-1}$ , for  $n \geq 1$ . 2
- (iii) In how many ways can a 2 by 5 grid be painted if 3 colours are available and each colour must now be used at least once? 2

**End of paper**

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# REFERENCE SHEET

- Mathematics –
- Mathematics Extension 1 –
- Mathematics Extension 2 –

# Mathematics

## Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

## Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

## Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

## Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

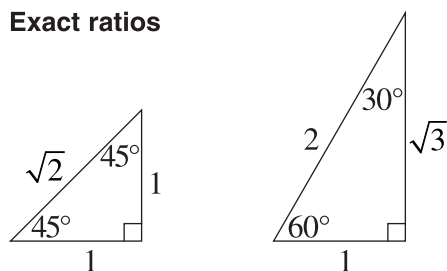
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

## Exact ratios



## Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

## Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

## Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

## Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

## Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

## Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

## Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

## $n$ th term of an arithmetic series

$$T_n = a + (n - 1)d$$

## Sum to $n$ terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

## $n$ th term of a geometric series

$$T_n = ar^{n-1}$$

## Sum to $n$ terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

## Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

## Compound interest

$$A_n = P \left( 1 + \frac{r}{100} \right)^n$$

# Mathematics (continued)

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## Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

## Derivatives

If  $y = x^n$ , then  $\frac{dy}{dx} = nx^{n-1}$

If  $y = uv$ , then  $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If  $y = \frac{u}{v}$ , then  $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If  $y = F(u)$ , then  $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If  $y = e^{f(x)}$ , then  $\frac{dy}{dx} = f'(x)e^{f(x)}$

If  $y = \log_e f(x) = \ln f(x)$ , then  $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If  $y = \sin f(x)$ , then  $\frac{dy}{dx} = f'(x) \cos f(x)$

If  $y = \cos f(x)$ , then  $\frac{dy}{dx} = -f'(x) \sin f(x)$

If  $y = \tan f(x)$ , then  $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

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## Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

## Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

## Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

## Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

## Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

## Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[ f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

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## Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

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## Angle measure

$$180^\circ = \pi \text{ radians}$$

## Length of an arc

$$l = r\theta$$

## Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

# Mathematics Extension 1

## Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

## t formulae

If  $t = \tan \frac{\theta}{2}$ , then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

## General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

## Division of an interval in a given ratio

$$\left( \frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

## Parametric representation of a parabola

For  $x^2 = 4ay$ ,

$$x = 2at, \quad y = at^2$$

At  $(2at, at^2)$ ,

$$\text{tangent: } y = tx - at^2$$

$$\text{normal: } x + ty = at^3 + 2at$$

At  $(x_1, y_1)$ ,

$$\text{tangent: } xx_1 = 2a(y + y_1)$$

$$\text{normal: } y - y_1 = -\frac{2a}{x_1}(x - x_1)$$

Chord of contact from  $(x_0, y_0)$ :  $xx_0 = 2a(y + y_0)$

## Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

## Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

## Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

## Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

## Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

## Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

## **2017 HSC Mathematics Extension 2**

### **Marking Guidelines**

#### **Section I**

##### **Multiple-choice Answer Key**

| <b>Question</b> | <b>Answer</b> |
|-----------------|---------------|
| 1               | C             |
| 2               | B             |
| 3               | D             |
| 4               | C             |
| 5               | B             |
| 6               | A             |
| 7               | A             |
| 8               | B             |
| 9               | C             |
| 10              | B             |

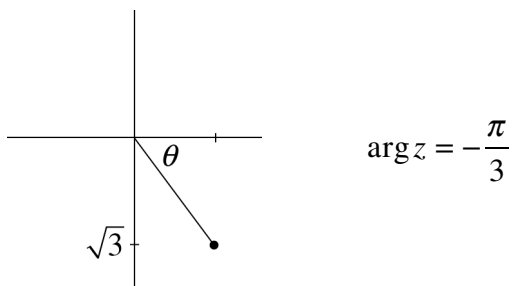
## Section II

### Question 11 (a) (i)

| Criteria                 | Marks |
|--------------------------|-------|
| • Provides correct value | 1     |

*Sample answer:*

$$z = 1 - i\sqrt{3}$$



### Question 11 (a) (ii)

| Criteria                                      | Marks |
|-----------------------------------------------|-------|
| • Provides correct solution                   | 2     |
| • Finds argument of $w$ , or equivalent merit | 1     |

*Sample answer:*

$$w = 1 + i$$

$$\arg w = \frac{\pi}{4}$$

$$\arg\left(\frac{z}{w}\right) = \arg z - \arg w$$

$$= \left(-\frac{\pi}{3}\right) - \left(\frac{\pi}{4}\right)$$

$$= \frac{-4\pi - 3\pi}{12}$$

$$= \frac{-7\pi}{12}$$

### Question 11 (b)

| Criteria                                                | Marks |
|---------------------------------------------------------|-------|
| • Provides correct solution                             | 2     |
| • Finds the slope of the asymptote, or equivalent merit | 1     |

*Sample answer:*

$$\frac{x^2}{12} - \frac{y^2}{4} = 1$$

$$a^2 = 12 \quad \therefore a = 2\sqrt{3}$$

$$b^2 = 4 \quad \therefore b = 2$$

$$\tan \alpha = \frac{b}{a} = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$$

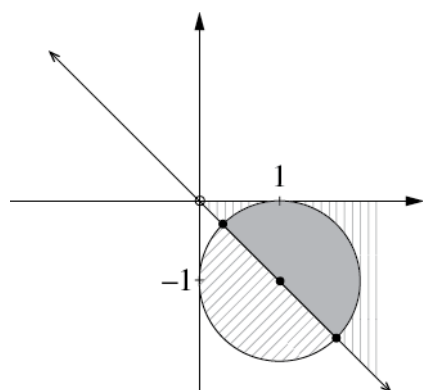
$$\alpha = \frac{\pi}{6}$$

### Question 11 (c)

| Criteria                                           | Marks |
|----------------------------------------------------|-------|
| • Provides correct sketch                          | 2     |
| • Sketches one correct region, or equivalent merit | 1     |

*Sample answer:*

$$-\frac{\pi}{4} \leq \arg z \leq 0 \quad \text{and} \quad |z - (1 - i)| \leq 1$$



**Question 11 (d)**

| Criteria                                               | Marks |
|--------------------------------------------------------|-------|
| • Provides correct solution                            | 3     |
| • Obtains correct primitive, or equivalent merit       | 2     |
| • Correctly deals with $d\theta$ , or equivalent merit | 1     |

**Sample answer:**

$$t = \tan \frac{\theta}{2}$$

$$\int_0^{\frac{2\pi}{3}} \frac{1}{1 + \cos \theta} \times d\theta = \int_0^{\sqrt{3}} \frac{\frac{2dt}{1+t^2}}{\frac{1+t^2}{1+t^2} + \frac{1-t^2}{1+t^2}}$$

$$= \int_0^{\sqrt{3}} \frac{2dt}{2}$$

$$= \int_0^{\sqrt{3}} 1 \times dt$$

$$= t \Big|_0^{\sqrt{3}}$$

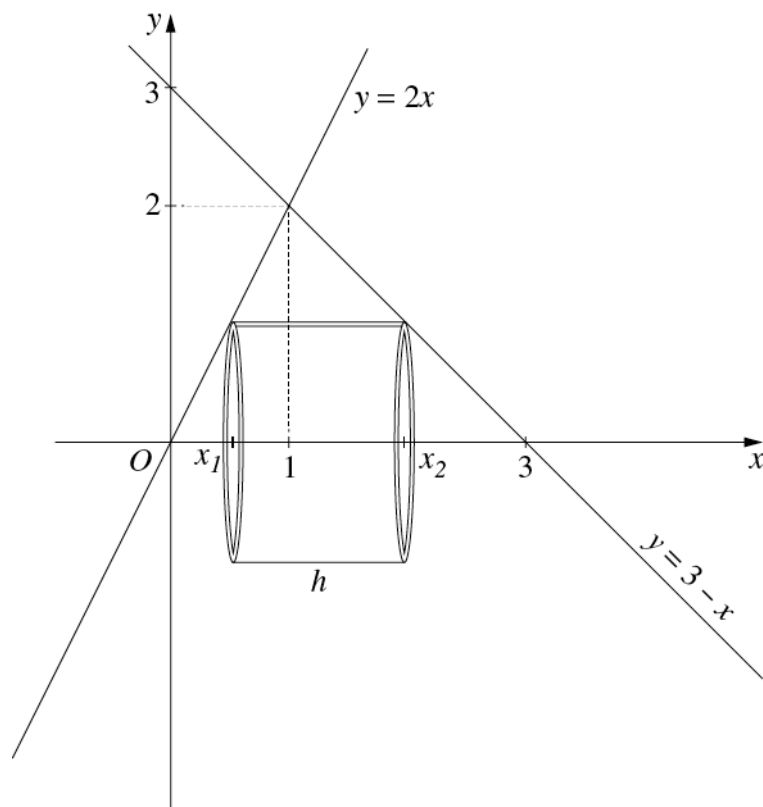
$$= \sqrt{3} - 0$$

$$= \sqrt{3}$$

**Question 11 (e)**

| Criteria                                                               | Marks |
|------------------------------------------------------------------------|-------|
| • Provides correct solution                                            | 2     |
| • Finds the height of the shells in terms of $y$ , or equivalent merit | 1     |

*Sample answer:*



$$\begin{aligned}
 r &= y \\
 h &= x_2 - x_1 \\
 &= (3 - y) - \left(\frac{y}{2}\right) \\
 &= 3 - \frac{3}{2}y
 \end{aligned}$$

$$\begin{aligned}
 V &= 2\pi \int_0^2 rh \, dy \\
 &= 2\pi \int_0^2 y \times \left(3 - \frac{3}{2}y\right) \times dy
 \end{aligned}$$

### Question 11 (f)

| Criteria                                                               | Marks |
|------------------------------------------------------------------------|-------|
| • Provides correct solution                                            | 3     |
| • Attempts to use a double angle result, or equivalent merit           | 2     |
| • Obtains correct integrand in terms of $\theta$ , or equivalent merit | 1     |

**Sample answer:**

$$x = \sin^2 \theta$$

$$dx = 2\sin\theta\cos\theta d\theta$$

$$\int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} \times dx = \int_0^{\frac{\pi}{4}} \sqrt{\frac{\sin^2\theta}{1-\sin^2\theta}} \times 2\sin\theta\cos\theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin\theta}{\cos\theta} \times 2\sin\theta\cos\theta d\theta$$

$$= 2 \int_0^{\frac{\pi}{4}} \sin^2\theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} (1 - \cos 2\theta) d\theta$$

$$= \theta - \frac{1}{2}\sin 2\theta \Bigg|_0^{\frac{\pi}{4}}$$

$$= \left( \frac{\pi}{4} - \frac{1}{2}\sin \frac{\pi}{2} \right) - (0)$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

**Question 12 (a) (i)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

$$f(x) = \frac{e^x - 1}{e^x + 1} \quad u = e^x - 1$$

$$u = e^x$$

$$v = e^x + 1$$

$$v = e^x$$

$$f'(x) = \frac{(e^x + 1)e^x - (e^x - 1)e^x}{(e^x + 1)^2}$$

$$= \frac{e^{2x} + e^x - e^{2x} + e^x}{(e^x + 1)^2}$$

$$= \frac{2e^x}{(e^x + 1)^2}$$

$$f'(x) > 0 \text{ for all } x \text{ as } 2e^x > 0 \text{ and } (e^x + 1)^2 > 0$$

$\therefore f(x)$  is increasing for all  $x$ .

**Question 12 (a) (ii)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

$$\begin{aligned}
 f(-x) &= \frac{e^{-x} - 1}{e^{-x} + 1} \\
 &= \frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1} \\
 &= \frac{\frac{1 - e^x}{e^x}}{\frac{1 + e^x}{e^x}} \\
 &= \frac{1 - e^x}{1 + e^x} \\
 &= \frac{-(-1 + e^x)}{e^x + 1} \\
 &= -\frac{(e^x - 1)}{e^x + 1} \\
 &= -[f(x)]
 \end{aligned}$$

$$\therefore f(-x) = -f(x)$$

the function is odd.

**Question 12 (a) (iii)**

| Criteria                       | Marks |
|--------------------------------|-------|
| • Provides correct description | 1     |

*Sample answer:*

$$\lim_{x \rightarrow \infty} \frac{e^x - 1}{e^x + 1} = 1$$

ie  $x \rightarrow \infty$ ,  $f(x) \rightarrow 1$  (from below)

**Question 12 (a) (iv)**

| Criteria                  | Marks |
|---------------------------|-------|
| • Provides correct sketch | 1     |

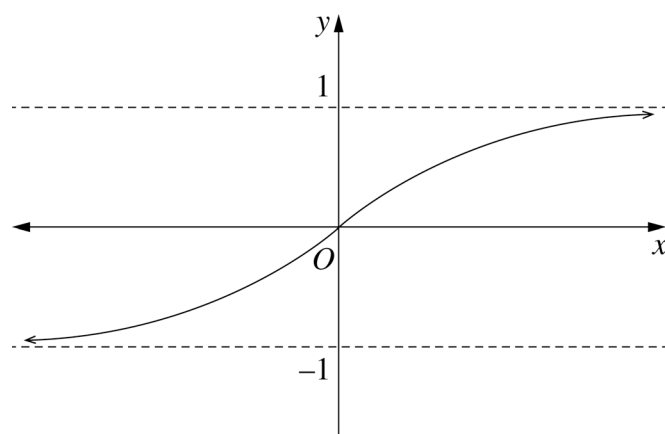
*Sample answer:*

$$f(0) = \frac{e^0 - 1}{e^0 + 1}$$

$$= \frac{1 - 1}{1 + 1}$$

$$= 0$$

$\therefore (0, 0)$  lies on the curve.



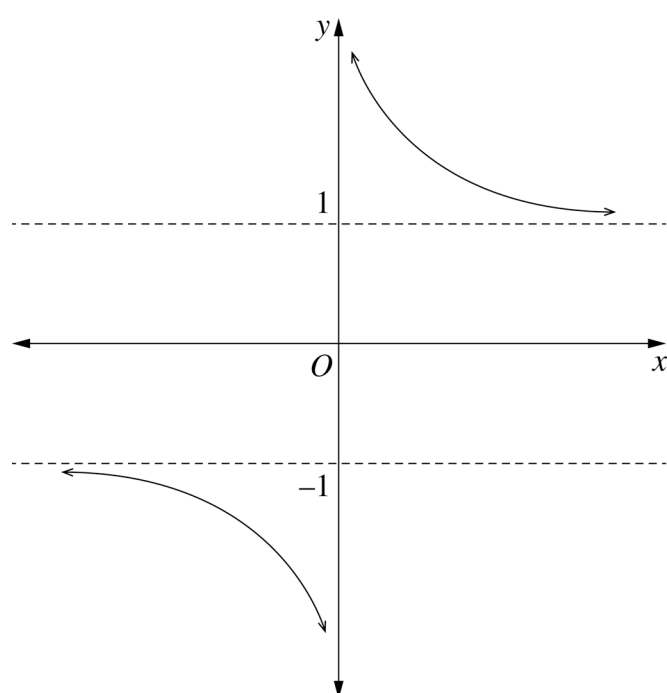
**Question 12 (a) (v)**

| Criteria                  | Marks |
|---------------------------|-------|
| • Provides correct sketch | 1     |

*Sample answer:*

$$y = \frac{e^x + 1}{e^x - 1}$$

$$y = \frac{1}{f(x)}$$



### Question 12 (b)

| Criteria                                                    | Marks |
|-------------------------------------------------------------|-------|
| • Provides correct solution                                 | 3     |
| • Correctly evaluates the discriminant, or equivalent merit | 2     |
| • Uses quadratic formula, or equivalent merit               | 1     |

**Sample answer:**

$$z^2 + (2 + 3i)z + (1 + 3i) = 0$$

$$a = 1$$

$$b = 2 + 3i$$

$$c = 1 + 3i$$

$$z = \frac{-(2 + 3i) \pm \sqrt{(2 + 3i)^2 - 4(1)(1 + 3i)}}{2(1)}$$

$$= \frac{-(2 + 3i) \pm \sqrt{4 + 12i + 9i^2 - 4 - 12i}}{2}$$

$$= \frac{-(2 + 3i) \pm \sqrt{9i^2}}{2}$$

$$= \frac{-(2 + 3i) \pm 3i}{2}$$

$$z = \frac{-2 - 3i + 3i}{2} \quad \text{and} \quad z = \frac{-2 - 3i - 3i}{2}$$

$$z = \frac{-2}{2} \quad \text{and} \quad z = \frac{-2 - 6i}{2}$$

$$z = -1 \quad \quad \quad z = -1 - 3i$$

### Question 12 (c)

| Criteria                                     | Marks |
|----------------------------------------------|-------|
| • Provides correct primitive                 | 3     |
| • Correctly uses parts, or equivalent merit  | 2     |
| • Attempts to use parts, or equivalent merit | 1     |

**Sample answer:**

$$I = \int x \tan^{-1} x \, dx$$

Using integration by parts,

$$u = \tan^{-1} x \quad dv = x dx$$

$$du = \frac{1}{1+x^2} dx \quad v = \frac{x^2}{2}$$

$$I = \frac{x^2}{2} \tan^{-1} x - \int \frac{x^2}{2} \left( \frac{1}{1+x^2} \right) dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2 + 1 - 1}{x^2 + 1} dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \left( 1 - \frac{1}{1+x^2} \right) dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left[ x - \tan^{-1} x \right] + c$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{x}{2} + \frac{\tan^{-1} x}{2} + c$$

$$= \frac{1}{2} (x^2 \tan^{-1} x - x + \tan^{-1} x) + c$$

**Question 12 (d) (i)**

| Criteria                                                             | Marks |
|----------------------------------------------------------------------|-------|
| • Provides correct solution                                          | 2     |
| • Recognises that $P(x) = (x - \alpha)^2 q(x)$ , or equivalent merit | 1     |

**Sample answer:**

$$\text{Let } P(x) = (x - \alpha)^2 \times Q(x)$$

$$P'(x) = 2(x - \alpha)' \times Q(x) + (x - \alpha)^2 \times Q'(x)$$

$$= (x - \alpha)[2 \times Q(x) + (x - \alpha) \times Q'(x)]$$

$$P(\alpha) = (\alpha - \alpha)^2 \times Q(\alpha)$$

$$= 0$$

$$P'(\alpha) = (\alpha - \alpha)[2 \times Q(\alpha) + (\alpha - \alpha) \times Q'(\alpha)]$$

$$= 0$$

**Question 12 (d) (ii)**

| Criteria                                             | Marks |
|------------------------------------------------------|-------|
| • Provides correct solution                          | 2     |
| • Attempt to solve $P'(x) = 0$ , or equivalent merit | 1     |

**Sample answer:**

$$P(x) = x^4 - 3x^3 + x^2 + 4$$

$$P'(x) = 4x^3 - 9x^2 + 2x$$

$$= x(4x^2 - 9x + 2)$$

$$= x(4x - 1)(x - 2)$$

Roots of  $P'(x)$  are  $x = 0, \frac{1}{4}, 2$

$$P(0) = 4$$

$$P\left(\frac{1}{4}\right) = \frac{1029}{256}$$

$$P(2) = 0$$

$\therefore (x - 2)^2$  is a factor of  $P(x)$

$$\alpha = 2$$

### Question 13 (a)

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

$$\text{As } (\sqrt{r} - \sqrt{s})^2 \geq 0$$

$$r - 2\sqrt{rs} + s \geq 0$$

$$\therefore \frac{r+s}{2} \geq \sqrt{rs}$$

### Question 13 (b) (i)

| Criteria                                                                      | Marks |
|-------------------------------------------------------------------------------|-------|
| • Provides correct proof                                                      | 2     |
| • Attempts to use sum of the threewise products of roots, or equivalent merit | 1     |

*Sample answer:*

$$-a = \text{sum of roots} = \alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta}$$

$$-c = \text{sum of products of 3 roots}$$

$$= \alpha \frac{1}{\alpha} \beta + \alpha \frac{1}{\alpha} \frac{1}{\beta} + \alpha \beta \frac{1}{\beta} + \frac{1}{\alpha} \beta \frac{1}{\beta}$$

$$= \beta + \frac{1}{\beta} + \alpha + \frac{1}{\alpha}$$

$$= -a$$

$$\therefore a = c$$

OR

The equation whose roots are reciprocals of the roots is found by replacing  $x$  by  $\frac{1}{x}$ .

$$\frac{1}{x^4} + a \frac{1}{x^3} + b \frac{1}{x^2} + c \frac{1}{x} + d$$

$$\text{ie } 1 + ax + bx^2 + cx^3 + dx^4$$

Since the reciprocals give the same set of roots we can match the coefficients so that  $d = 1$  and  $c = a$ .

**Question 13 (b) (ii)**

| Criteria                                              | Marks |
|-------------------------------------------------------|-------|
| • Provides correct solution                           | 2     |
| • Uses sum of the roots in pairs, or equivalent merit | 1     |

*Sample answer:*

$b$  = sum of pairs of roots

$$= \alpha \frac{1}{\alpha} + \alpha\beta + \alpha \frac{1}{\beta} + \beta \frac{1}{\alpha} + \frac{1}{\alpha} \frac{1}{\beta} + \beta \frac{1}{\beta}$$

$$= 1 + \alpha\beta + \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + \frac{1}{\alpha\beta} + 1$$

$$= 1 + \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + \alpha\beta + \frac{1}{\alpha\beta}$$

Now  $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} \geq \frac{2\sqrt{\alpha^2\beta^2}}{\alpha\beta}$  from 13 part (a)

$$\therefore \frac{\alpha}{\beta} + \frac{\beta}{\alpha} \geq 2 \quad \text{and similarly} \quad \alpha\beta + \frac{1}{\alpha\beta} \geq 2$$

$$\therefore b \geq 2 + 2 + 2 = 6$$

### Question 13 (c)

| Criteria                                                                                 | Marks |
|------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                              | 4     |
| • Obtains correct expression for $x$ in terms of $v$ , $k$ and $g$ , or equivalent merit | 3     |
| • Separates the variables, or equivalent merit                                           | 2     |
| • Rewrites acceleration as $v \frac{dv}{dx}$ , or equivalent merit                       | 1     |

**Sample answer:**

$$\ddot{x} = v \frac{dv}{dx} = -g - kv^2$$

$$\therefore \frac{dv}{dx} = -\frac{g + kv^2}{v}$$

$$\frac{dx}{dv} = -\frac{v}{g + kv^2}$$

$$\therefore x = -\int \frac{v}{g + kv^2} dv$$

$$= -\frac{1}{2k} \log_e(g + kv^2) + c$$

when  $x = 0$   $v = \frac{1}{2} \sqrt{\frac{g}{k}}$  so that  $v^2 = \frac{g}{4k}$

hence  $0 = -\frac{1}{2k} \log_e\left(g + \frac{kg}{4k}\right) + c$

$$\therefore c = \frac{1}{2k} \log_e\left(\frac{5g}{4}\right)$$

$$\therefore x = \frac{1}{2k} \log_e\left(\frac{5g}{4}\right) - \frac{1}{2k} \log_e(g + kv^2)$$

Maximum height occurs when  $v = 0$

$$\therefore H = \frac{1}{2k} \left[ \log_e\left(\frac{5g}{4}\right) - \log_e(g) \right]$$

$$= \frac{1}{2k} \log_e\left(\frac{5}{4}\right)$$

### Question 13 (d)

| Criteria                                                                    | Marks |
|-----------------------------------------------------------------------------|-------|
| • Provides correct solution                                                 | 4     |
| • Obtains an integral for the volume, or equivalent merit                   | 3     |
| • Finds the area of the cross-section in terms of $x$ , or equivalent merit | 2     |
| • Finds the height of $\triangle PQR$ in terms of $x$ , or equivalent merit | 1     |

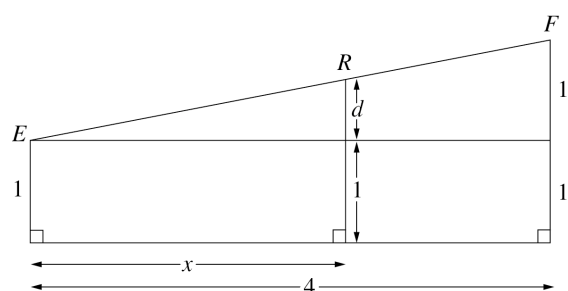
**Sample answer:**

The line  $BC$  has equation  $y = \left(\frac{3-1}{4-0}\right)x + 1$

$$y = \frac{1}{2}x + 1$$

$$\therefore QP = 2\left(\frac{1}{2}x + 1\right) = x + 2$$

To find height of  $\triangle QRP$



$$\therefore d = \frac{x}{4}$$

$$\therefore \text{height} = 1 + d = \frac{x}{4} + 1$$

By similar triangles  $\frac{d}{x} = \frac{1}{4}$

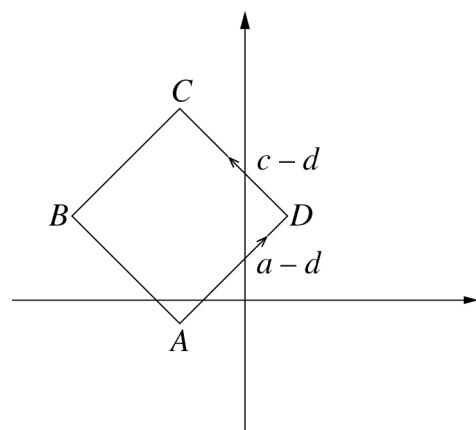
$$\begin{aligned}
 \therefore \text{Area } \triangle QRP &= \frac{1}{2}QP\left(\frac{x}{4}+1\right) \\
 &= \left(\frac{x}{2}+1\right)\left(\frac{x}{4}+1\right) \\
 &= \frac{x^2}{8} + \frac{3x}{4} + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{Volume} &= \frac{1}{8} \int_0^4 x^2 + 6x + 8 \, dx \\
 &= \frac{1}{8} \left[ \frac{x^3}{3} + 3x^2 + 8x \right]_0^4 \\
 &= \frac{1}{8} \left[ \frac{64}{3} + 48 + 32 - 0 \right] \\
 &= \frac{38}{3} \text{ units}^3
 \end{aligned}$$

### Question 13 (e)

| Criteria                                           | Marks |
|----------------------------------------------------|-------|
| • Provides correct solution                        | 2     |
| • Obtains $c - d = i(d - a)$ , or equivalent merit | 1     |

*Sample answer:*



$$c - d = i(d - a)$$

(rotation by  $\frac{\pi}{2}$ )

$$c = d + id - ia$$

$$= (1 + i)d - ia$$

**Question 14 (a) (i)**

| Criteria                  | Marks |
|---------------------------|-------|
| • Provides correct values | 1     |

*Sample answer:*

Multiply both sides by  $x^4 + 4$

$$16 = (A + 2x)(x^2 - 2x + 2) + (B - 2x)(x^2 + 2x + 2)$$

$$\text{at } x = 0 \quad 16 = 2A + 2B \quad A + B = 8 \quad \text{--- ①}$$

$$\begin{aligned} \text{at } x = 1 \quad 16 &= (A + 2) + (B - 2)5 \\ 16 &= A + 2 + 5B - 10 \quad A + 5B = 24 \quad \text{--- ②} \end{aligned}$$

Subtract ② – ①

$$4B = 16$$

$$B = 4$$

$$A = 4$$

**Question 14 (a) (ii)**

| Criteria                                         | Marks |
|--------------------------------------------------|-------|
| • Provides correct solution                      | 2     |
| • Obtains correct primitive, or equivalent merit | 1     |

*Sample answer:*

$$\begin{aligned}
 & \int_0^m \frac{16}{x^4 + 4} dx \\
 &= \int_0^m \frac{4 + 2x}{x^2 + 2x + 2} + \int_0^m \frac{4 - 2x}{x^2 - 2x + 2} dx \\
 &= \int_0^m \frac{2}{x^2 + 2x + 2} + \frac{2x + 2}{x^2 + 2x + 2} dx + \int_0^m \frac{2}{x^2 - 2x + 2} - \frac{2x - 2}{x^2 - 2x + 2} dx \\
 &= \int_0^m \frac{2}{1 + (x + 1)^2} dx + \left[ \ln(x^2 + 2x + 2) \right]_0^m + \int_0^m \frac{2}{1 + (x - 1)^2} dx - \left[ \ln(x^2 - 2x + 2) \right]_0^m \\
 &= \left[ 2 \tan^{-1}(x + 1) \right]_0^m + \ln(m^2 + 2m + 2) - \log 2 + \left[ 2 \tan^{-1}(x - 1) \right]_0^m - \ln(m^2 - 2m + 2) + \log 2 \\
 &= 2 \tan^{-1}(m + 1) - \frac{\pi}{2} + \ln\left(\frac{m^2 + 2m + 2}{m^2 - 2m + 2}\right) + 2 \tan^{-1}(m - 1) + \frac{\pi}{2} \\
 &= \ln\left(\frac{m^2 + 2m + 2}{m^2 - 2m + 2}\right) + 2 \tan^{-1}(m + 1) + 2 \tan^{-1}(m - 1)
 \end{aligned}$$

**Question 14 (a) (iii)**

| Criteria                  | Marks |
|---------------------------|-------|
| • Provides correct answer | 1     |

*Sample answer:*

$$\int_0^m \frac{16}{x^4 + 4} dx = \ln \left( \frac{1 + \frac{2}{m} + \frac{2}{m^2}}{1 - \frac{2}{m} + \frac{2}{m^2}} \right) + 2 \tan^{-1}(m+1) + 2 \tan^{-1}(m-1)$$

as  $m \rightarrow \infty$

$$\rightarrow \ln(1) + 2\left(\frac{\pi}{2}\right) + 2\left(\frac{\pi}{2}\right) = 2\pi$$

**Question 14 (b) (i)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

Construct  $DB$

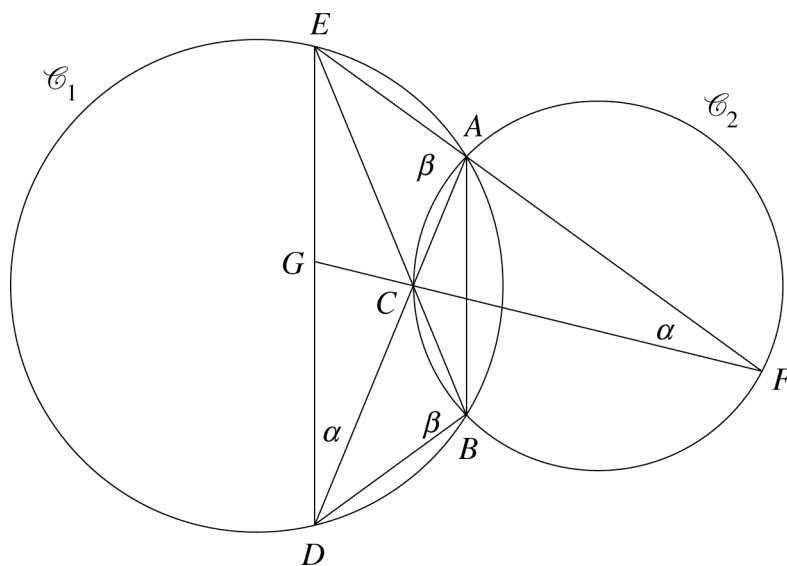
$\angle EAD = \angle EBD$ , as they stand on the same arc,  $ED$  of  $C_1$ .

**Question 14 (b) (ii)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

Construct  $AB$



$\angle EDA = \angle EBA$  angles standing on arc  $EA$  of  $C_1$ .

$\angle EBA = \angle CBA$   
 $= \angle CFA$  angles standing on arc  $AC$  in other circle  $C_2$ .

so  $\angle EDA = \angle EBA = \angle CFA = \angle AFC$

**Question 14 (b) (iii)**

| Criteria                                                  | Marks |
|-----------------------------------------------------------|-------|
| • Provides correct solution                               | 3     |
| • Makes substantial progress, or equivalent merit         | 2     |
| • Attempts to find an exterior angle, or equivalent merit | 1     |

**Sample answer:**

Let  $\angle AFC = \angle EDA = \alpha$   
 $\angle EAD = \angle EBD = \beta$

In  $\triangle EAD$ ,  $\angle AED = \pi - \alpha - \beta$

In  $\triangle EFG$ ,  $\angle EGF = \pi - (\alpha + (\pi - \alpha - \beta))$   
 $= \beta$

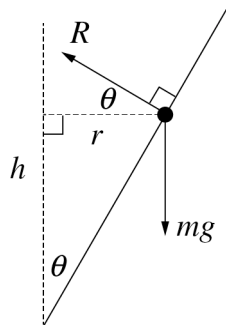
so  $\angle EGF = \angle EBD = \beta$

Exterior angle ( $EGF$ ) equals opposite interior angle ( $EBD$ ) and so  $BCGD$  is a cyclic quadrilateral.

**Question 14 (c) (i)**

| Criteria                                                     | Marks |
|--------------------------------------------------------------|-------|
| • Provides correct solution                                  | 2     |
| • Obtains one correct component of $R$ , or equivalent merit | 1     |

*Sample answer:*



$$R \sin \theta = mg$$

$$R \cos \theta = mrw^2$$

Dividing

$$\tan \theta = \frac{g}{rw^2}$$

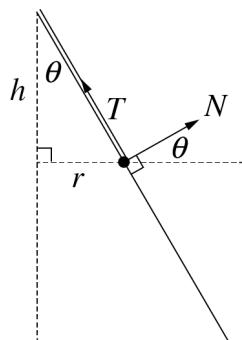
$$\frac{r}{h} = \frac{g}{rw^2}$$

$$w^2 = \frac{gh}{r^2}$$

**Question 14 (c) (ii)**

| Criteria                                       | Marks |
|------------------------------------------------|-------|
| • Provides correct solution                    | 2     |
| • Correctly resolves $N$ , or equivalent merit | 1     |

*Sample answer:*



Horizontal:

$$T \sin \theta - N \cos \theta = mrw^2$$

$$T \sin \theta \cos \theta - N \cos^2 \theta = mrw^2 \cos \theta \quad \text{--- ①}$$

Vertical:

$$T \cos \theta + N \sin \theta = mg$$

$$T \sin \theta \cos \theta + N \sin^2 \theta = mg \sin \theta \quad \text{--- ②}$$

$$\text{②} - \text{①} \quad N(\sin^2 \theta + \cos^2 \theta) = mg \sin \theta - mrw^2 \cos \theta$$

**Question 14 (c) (iii)**

| Criteria                                    | Marks |
|---------------------------------------------|-------|
| • Provides correct solution                 | 2     |
| • Observes $N \geq 0$ , or equivalent merit | 1     |

***Sample answer:***

As particle 2 moves in a circle of radius  $r$  it must have

$$N \geq 0$$

$$mg\left(\sin\theta - \frac{h}{r}\cos\theta\right) \geq 0$$

$$\sin\theta - \cot\theta\cos\theta \geq 0$$

$$\sin\theta \geq \cot\theta\cos\theta$$

$$\tan\theta \geq \cot\theta$$

In first quad so

$$\theta \geq \frac{\pi}{4}$$

**Question 15 (a) (i)**

| Criteria                  | Marks |
|---------------------------|-------|
| • Provides correct answer | 1     |

*Sample answer:*

$$\begin{aligned}
 I_1 &= \int_0^1 x^1 \sqrt{1-x^2} \, dx \\
 &= \frac{1}{2} \int_0^1 -2x \sqrt{1-x^2} \, dx \\
 &= -\frac{1}{2} \left[ \frac{2}{3} (1-x^2)^{\frac{3}{2}} \right]_0^1 \\
 &= -\frac{1}{3} (0-1) \\
 &= \frac{1}{3}
 \end{aligned}$$

**Question 15 (a) (ii)**

| Criteria                                                                               | Marks |
|----------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                            | 3     |
| • Correctly uses integration by parts, simplifying where possible, or equivalent merit | 2     |
| • Attempts to use integration by parts, or equivalent merit                            | 1     |

*Sample answer:*

$$I_n = \int_0^1 x^n \sqrt{1-x^2} dx$$

$$\text{Let } u = x^{n-1} \quad dv = x\sqrt{1-x^2} dx$$

$$du = (n-1)x^{n-2} dx \quad v = -\frac{1}{3}(1-x^2)^{\frac{3}{2}}$$

$$\therefore I_n = \left[ x^{n-1} \times \frac{-1}{3}(1-x^2)^{\frac{3}{2}} \right]_0^1 + \frac{1}{3} \int_0^1 (1-x^2)^{\frac{3}{2}} (n-1)x^{n-2} dx$$

$$= \frac{(n-1)}{3} \int_0^1 x^{n-2} (1-x^2) \sqrt{1-x^2} dx$$

$$= \frac{(n-1)}{3} \int_0^1 x^{n-2} \sqrt{1-x^2} dx - \frac{n-1}{3} \int_0^1 x^n \sqrt{1-x^2} dx$$

$$\therefore I_n = \frac{n-1}{3} I_{n-2} - \frac{n-1}{3} I_n$$

$$I_n \left( 1 + \frac{n-1}{3} \right) = \frac{n-1}{3} I_{n-2}$$

$$I_n \left( \frac{3+n-1}{3} \right) = \frac{n-1}{3} I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n+2} I_{n-2}$$

**Question 15 (a) (iii)**

| Criteria                  | Marks |
|---------------------------|-------|
| • Provides correct answer | 1     |

*Sample answer:*

$$\begin{aligned}I_5 &= \frac{4}{7} \times I_3 \\&= \frac{4}{7} \times \frac{2}{5} \times I_1 \\&= \frac{4}{7} \times \frac{2}{5} \times \frac{1}{3} \quad \text{from part (i)} \\&= \frac{8}{105}\end{aligned}$$

**Question 15 (b) (i)**

| Criteria                                                   | Marks |
|------------------------------------------------------------|-------|
| • Provides correct solution                                | 2     |
| • Differentiates with respect to $x$ , or equivalent merit | 1     |

**Sample answer:**

$$x^{\frac{1}{2}} + y^{\frac{1}{2}} = a^{\frac{1}{2}} \quad (a \text{ constant})$$

Differentiating implicitly:

$$\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}y^{\frac{1}{2}} \times \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{x^{-\frac{1}{2}}}{y^{-\frac{1}{2}}}$$

$$= -\frac{y^{\frac{1}{2}}}{x^{\frac{1}{2}}}$$

$$= -\sqrt{\frac{y}{x}}$$

$$\therefore \text{At } P, \text{ gradient of tangent} = -\sqrt{\frac{d}{c}}$$

and equation of tangent is

$$y - d = -\sqrt{\frac{d}{c}}(x - c)$$

$$\sqrt{c}y - \sqrt{c}d = -\sqrt{d}x + \sqrt{d}c$$

$$\therefore y\sqrt{c} + x\sqrt{d} = d\sqrt{c} + c\sqrt{d}$$

**Question 15 (b) (ii)**

| Criteria                                                                            | Marks |
|-------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                         | 3     |
| • Obtains an expression for $OA + OB$ in terms of $c$ and $d$ , or equivalent merit | 2     |
| • Finds an expression for $OA$ , or equivalent merit                                | 1     |

**Sample answer:**

At  $A$ ,  $y = 0$

$$\therefore x\sqrt{d} = d\sqrt{c} + c\sqrt{d}$$

$$x = \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{d}}$$

$$\therefore OA = \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{d}}$$

At  $B$ ,  $x = 0$

$$\therefore y\sqrt{c} = d\sqrt{c} + c\sqrt{d}$$

$$\therefore OB = \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{c}}$$

$$\begin{aligned} \therefore OA + OB &= \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{d}} + \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{c}} \\ &= \sqrt{d}\sqrt{c} + c + d + \sqrt{c}\sqrt{d} \\ &= \sqrt{c}(\sqrt{d} + \sqrt{c}) + \sqrt{d}(\sqrt{d} + \sqrt{c}) \\ &= (\sqrt{c} + \sqrt{d})(\sqrt{d} + \sqrt{c}) \end{aligned}$$

But  $P$  lies on  $\sqrt{x} + \sqrt{y} = \sqrt{a}$

$$\therefore \sqrt{c} + \sqrt{d} = \sqrt{a}$$

$$\begin{aligned} \therefore OA + OB &= \sqrt{a} \times \sqrt{a} \\ &= a \end{aligned}$$

**Question 15 (c) (i)**

| Criteria                                                            | Marks |
|---------------------------------------------------------------------|-------|
| • Provides correct solution                                         | 2     |
| • Attempts to solve the simultaneous equations, or equivalent merit | 1     |

**Sample answer:**

$$\text{At } P, \quad \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1 \quad \text{and} \quad \frac{x_1^2}{c^2} - \frac{y_1^2}{d^2} = 1$$

$$\therefore \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = \frac{x_1^2}{c^2} - \frac{y_1^2}{d^2}$$

$$x_1^2 \left( \frac{1}{a^2} - \frac{1}{c^2} \right) = -y_1^2 \left( \frac{1}{d^2} + \frac{1}{b^2} \right)$$

$$x_1^2 \left( \frac{c^2 - a^2}{a^2 c^2} \right) = -y_1^2 \left( \frac{b^2 + d^2}{b^2 d^2} \right)$$

$$\therefore \frac{x_1^2}{y_1^2} = - \left( \frac{b^2 + d^2}{b^2 d^2} \right) \left( \frac{a^2 c^2}{c^2 - a^2} \right)$$

$$= \frac{a^2 c^2}{(a^2 - c^2)} \times \frac{(b^2 + d^2)}{b^2 d^2}$$

**Question 15 (c) (ii)**

| Criteria                                                 | Marks |
|----------------------------------------------------------|-------|
| • Provides correct solution                              | 3     |
| • Finds product of the two slopes, or equivalent merit   | 2     |
| • Finds slope of tangent to ellipse, or equivalent merit | 1     |

**Sample answer:**

Ellipse:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\therefore \frac{2x}{a^2} + \frac{2y}{b^2} \times \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{a^2} \times \frac{b^2}{y}$$

$$\therefore \text{Gradient of tangent at } P = \frac{-x_1 b^2}{a^2 y_1}$$

Hyperbola:  $\frac{x^2}{c^2} - \frac{y^2}{d^2} = 1$

Similarly gradient of tangent at  $P = \frac{x_1}{c^2} \times \frac{d^2}{y_1}$

$$\begin{aligned} \therefore \text{Product of gradients} &= \frac{-x_1 b^2}{a^2 y_1} \times \frac{x_1 d^2}{c^2 y_1} \\ &= \frac{-x_1^2}{y_1^2} \times \frac{b^2 d^2}{a^2 c^2} \\ &= \frac{-a^2 c^2}{(a^2 - c^2)} \times \frac{(b^2 + d^2)}{b^2 d^2} \times \frac{b^2 d^2}{a^2 c^2} \\ &= \frac{-(b^2 + d^2)}{a^2 - c^2} \end{aligned}$$

from part (i)

But, conics have same foci so  $ae = cE$

$$\begin{aligned} \text{and } b^2 &= a^2(1 - e^2) & \text{and } d^2 &= c^2(E^2 - 1) \\ b^2 &= a^2 - a^2 e^2 & d^2 &= c^2 E^2 - c^2 \end{aligned}$$

$$\begin{aligned} \therefore b^2 + d^2 &= a^2 - a^2 e^2 + c^2 E^2 - c^2 \\ &= a^2 - a^2 e^2 + a^2 e^2 - c^2 \\ &= a^2 - c^2 \end{aligned}$$

$$\begin{aligned}\therefore \text{Product of gradients} &= \frac{-(b^2 + d^2)}{a^2 - c^2} \\ &= \frac{-(a^2 - c^2)}{a^2 - c^2} \\ &= -1\end{aligned}$$

**Question 16 (a) (i)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

$$\alpha = \cos \theta + i \sin \theta$$

By de Moivre's theorem,

$$\alpha^k = \cos(k\theta) + i \sin(k\theta)$$

$$\alpha^{-k} = \cos(-k\theta) + i \sin(-k\theta)$$

$$= \cos(k\theta) - i \sin(k\theta)$$

$$\therefore \alpha^k + \alpha^{-k} = 2 \cos k\theta$$

**Question 16 (a) (ii)**

| Criteria                                                                                                         | Marks |
|------------------------------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                                                      | 3     |
| • Sums the series, multiplies top and bottom by $(1 - \bar{\alpha})$ and attempts to expand, or equivalent merit | 2     |
| • Recognises that there are $2n + 1$ terms, or equivalent merit                                                  | 1     |

*Sample answer:*

There are  $(2n + 1)$  terms in the geometric series with first term  $\alpha^{-n}$  and ratio  $\alpha$

$$\begin{aligned}
 C &= \frac{\alpha^{-n}(1 - \alpha^{2n+1})}{(1 - \alpha)} \\
 &= \frac{\alpha^{-n}(1 - \alpha^{2n+1})}{1 - \alpha} \times \left( \frac{1 - \bar{\alpha}}{1 - \bar{\alpha}} \right) \\
 &= \frac{\alpha^{-n} - \alpha^{-(n+1)} - \alpha^{(n+1)} + \alpha^n}{(1 - \alpha)(1 - \bar{\alpha})}, \quad \text{using } \bar{\alpha} = \frac{1}{\alpha} \\
 &= \frac{(\alpha^n + \alpha^{-n}) - (\alpha^{(n+1)} + \alpha^{-(n+1)})}{(1 - \alpha)(1 - \bar{\alpha})}
 \end{aligned}$$

**Question 16 (a) (iii)**

| Criteria                                                                                                 | Marks |
|----------------------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                                              | 2     |
| • Correctly matches appropriate terms and attempts to use the result from part (i) , or equivalent merit | 1     |

**Sample answer:**

Write  $C$  as

$$\begin{aligned}
 C &= 1 + (\alpha + \alpha^{-1}) + (\alpha^2 + \alpha^{-2}) + \cdots + (\alpha^n + \alpha^{-n}) \\
 &= 1 + 2\cos\theta + 2\cos 2\theta + \cdots + 2\cos(n\theta) \quad \text{by part (i)} \\
 &= 1 + 2(\cos\theta + \cos 2\theta + \cdots + \cos(n\theta))
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } C &= \frac{\alpha^n + \alpha^{-n} - (\alpha^{n+1} + \alpha^{-(n+1)})}{(1 - \alpha)(1 - \bar{\alpha})} \\
 &= \frac{2\cos n\theta - 2\cos(n+1)\theta}{2 - (\alpha + \bar{\alpha})} \\
 &= \frac{2\cos n\theta - 2\cos(n+1)\theta}{2 - 2\cos\theta} \\
 &= \frac{\cos n\theta - \cos(n+1)\theta}{1 - \cos\theta}
 \end{aligned}$$

**Question 16 (a) (iv)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

Put  $\theta = \frac{\pi}{n}$ , then

$$1 + 2 \left( \cos \frac{\pi}{n} + \cos \frac{2\pi}{n} + \cdots + \cos \frac{n\pi}{n} \right) = \frac{\cos \left( \frac{n\pi}{n} \right) - \cos \left( (n+1) \frac{\pi}{n} \right)}{1 - \cos \frac{\pi}{n}}$$

$$RHS = \frac{-1 - \cos \left( \pi + \frac{\pi}{n} \right)}{1 - \cos \frac{\pi}{n}}$$

$$= \frac{-1 + \cos \frac{\pi}{n}}{1 - \cos \frac{\pi}{n}}$$

$$= -1$$

$$\therefore LHS = 1 + 2 \left( \cos \frac{\pi}{n} + \cos \frac{2\pi}{n} + \cdots + \cos \frac{n\pi}{n} \right) = -1$$

$$\therefore \cos \frac{\pi}{n} + \cos \frac{2\pi}{n} + \cdots + \cos \left( \frac{n\pi}{n} \right) = -1$$

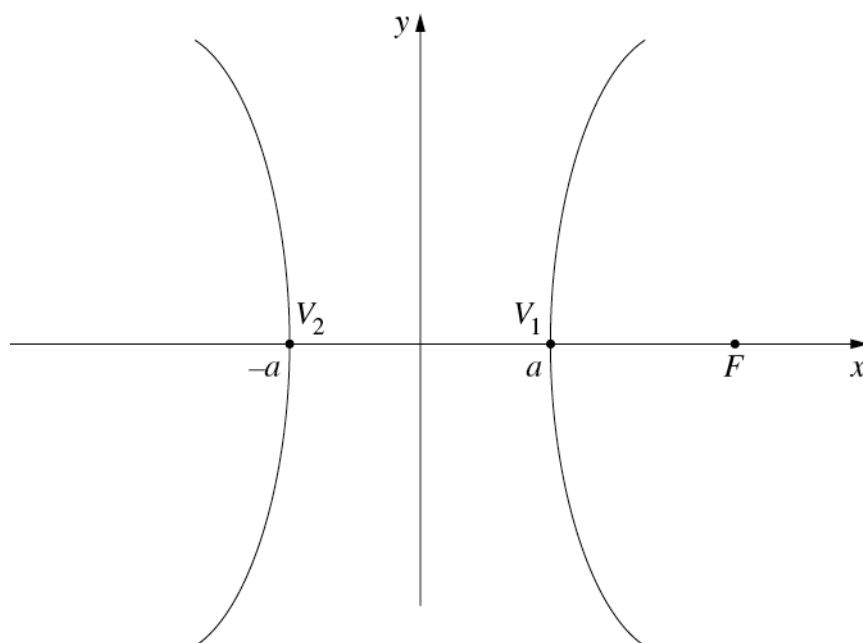
Which is independent of  $n$ .

**Question 16 (b)**

| Criteria                                               | Marks |
|--------------------------------------------------------|-------|
| • Provides correct solution                            | 2     |
| • Finds one correct value of $a$ , or equivalent merit | 1     |

**Sample answer:**

Since the eccentricity is 2 the conic is a hyperbola



The focus  $F$  is  $(ae, 0) = (2a, 0)$

$$V_1F = 1, \text{ so } 2a - a = 1 \Rightarrow a = 1$$

$$FV_2 = 2a + a = 1 \Rightarrow a = \frac{1}{3}$$

**Question 16 (c) (i)**

| Criteria                                                  | Marks |
|-----------------------------------------------------------|-------|
| • Provides correct solution                               | 2     |
| • Obtains correct count for one case, or equivalent merit | 1     |

**Sample answer:**

|   |   |
|---|---|
| A | C |
| B | D |

Case 1: Tiles  $B$  and  $C$  have the same colour.  
There are  $(x - 1)$  ways of choosing tile  $D$ .

Case 2: Tiles  $B$  and  $C$  have different colours.  
There are  $(x - 2)$  choices of colour for tile  $C$   
and there are  $(x - 2)$  choices of colour for tile  $D$   
giving  $(x - 2)^2$  choices in total.

$\therefore$  There are  $(x - 2)^2 + (x - 1) = x^2 - 3x + 3$  choices in total.

**Question 16 (c) (ii)**

| Criteria                        | Marks |
|---------------------------------|-------|
| • Provides correct proof        | 2     |
| • Demonstrates true for $n = 1$ | 1     |

**Sample answer:**

Let  $P(n)$  be the given proposition.

$P(1)$  is true since this was given.

Let  $k$  be an integer for which  $P(k)$  is true.

That is, there are  $x(x - 1)(x^2 - 3x + 3)^{k-1}$  choices for the colours. Consider a  $(k + 1) \times 2$  grid.

|  |   |   |
|--|---|---|
|  | A | C |
|  | B | D |

$\longleftarrow k \longrightarrow$

As in part (i) there are  $(x^2 - 3x + 3)$  choices for the colours of tiles  $C$  and  $D$ .

$\therefore$  There are  $x(x - 1)(x^2 - 3x + 3)^k$  choices for the colours.

$\therefore P(k + 1)$  is true, so  $P(n)$  is true for all  $n \geq 1$  by induction.

**Question 16 (c) (iii)**

| Criteria                                                                        | Marks |
|---------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                     | 2     |
| • Finds the number of ways if not all colours need be used, or equivalent merit | 1     |

***Sample answer:***

$$\text{Let } G(x) = x(x-1)(x^2-3x+3)^4$$

The number of ways to colour the grid with 3 colours, each used at least once, is

$$G(3) - 3 \times G(2) \quad (\text{there are 3 choices for which colour is left out})$$

$$= 6 \times 3^4 - 3 \times 2 \times 1^4$$

$$= 480$$

# 2017 HSC Mathematics Extension 2

## Mapping Grid

### Section I

| Question | Marks | Content | Syllabus outcomes |
|----------|-------|---------|-------------------|
| 1        | 1     | 2.4     | E3                |
| 2        | 1     | 3.4     | E3                |
| 3        | 1     | 2.2     | E3                |
| 4        | 1     | 1       | E6                |
| 5        | 1     | 7.5     | E4                |
| 6        | 1     | 7.5     | E3                |
| 7        | 1     | 1.9     | E6                |
| 8        | 1     | 1       | E9                |
| 9        | 1     | 6.3.1   | E5                |
| 10       | 1     | 8       | E6                |

### Section II

| Question     | Marks | Content | Syllabus outcomes |
|--------------|-------|---------|-------------------|
| 11 (a) (i)   | 1     | 2.2     | E3                |
| 11 (a) (ii)  | 2     | 2.2     | E3                |
| 11 (b)       | 2     | 3.2     | E3                |
| 11 (c)       | 2     | 2.5     | E3                |
| 11 (d)       | 3     | 4.1     | E8                |
| 11 (e)       | 2     | 5.1     | E7                |
| 11 (f)       | 3     | 4.1     | E8                |
| 12 (a) (i)   | 1     | 8       | E6                |
| 12 (a) (ii)  | 1     | 8       | E6                |
| 12 (a) (iii) | 1     | 8       | E6                |
| 12 (a) (iv)  | 1     | 8       | E6                |
| 12 (a) (v)   | 1     | 1.5     | E6                |
| 12 (b)       | 3     | 2.1     | E3                |
| 12 (c)       | 3     | 4.1     | E8                |
| 12 (d) (i)   | 2     | 7.2     | E4                |
| 12 (d) (ii)  | 2     | 7.2     | E4                |
| 13 (a)       | 1     | 8.3     | E2                |
| 13 (b) (i)   | 2     | 7.5     | E4                |
| 13 (b) (ii)  | 2     | 8.3     | E4                |

| Question     | Marks | Content      | Syllabus outcomes |
|--------------|-------|--------------|-------------------|
| 13 (c)       | 4     | 6.2.2        | E5                |
| 13 (d)       | 4     | 5.1          | E7                |
| 13 (e)       | 2     | 2.3          | E3                |
| 14 (a) (i)   | 1     | 7.6          | E8                |
| 14 (a) (ii)  | 2     | 4.1          | E8                |
| 14 (a) (iii) | 1     | 4.1          | E9                |
| 14 (b) (i)   | 1     | 8.1(2.9)     | E2                |
| 14 (b) (ii)  | 1     | 8.1(2.9)     | E2                |
| 14 (b) (iii) | 3     | 8.1(2.9)     | E2                |
| 14 (c) (i)   | 2     | 6.3.4        | E5                |
| 14 (c) (ii)  | 2     | 6.3.3        | E5                |
| 14 (c) (iii) | 2     | 6.3.3, 6.3.4 | E5                |
| 15 (a) (i)   | 1     | 4.1          | E8                |
| 15 (a) (ii)  | 3     | 4.1          | E8                |
| 15 (a) (iii) | 1     | 4.1          | E8                |
| 15 (b) (i)   | 2     | 1.8          | E6                |
| 15 (b) (ii)  | 3     | 8            | E6                |
| 15 (c) (i)   | 2     | 3.1, 3.2     | E4                |
| 15 (c) (ii)  | 3     | 3.1, 3.2     | E4                |
| 16 (a) (i)   | 1     | 2.4          | E3                |
| 16 (a) (ii)  | 3     | 2.1, 8       | E3                |
| 16 (a) (iii) | 2     | 2.1, 8       | E3                |
| 16 (a) (iv)  | 1     | 8            | E3                |
| 16 (b)       | 2     | 3.2          | E3                |
| 16 (c) (i)   | 2     | 8            | E2                |
| 16 (c) (ii)  | 2     | 8.2          | E2                |
| 16 (c) (iii) | 2     | 8            | E2                |